Impact of Fuel Prices and Metro Fares on Public Transport Usage in Istanbul

Final Report (revised) · DSA 210 · Elif Naz Sarıbaş · Sabancı University · October 2026

Summary

This project asks whether fuel prices, metro fares and car sales explain annual metro ridership in Istanbul between 2013 and 2024. A first analysis with nominal (unadjusted) prices found strong positive correlations of fuel prices (r = 0.79) and fares (r = 0.79) with ridership. This revised report shows that those correlations are largely an artifact of inflation: nominal prices rose roughly ninefold while ridership also trended upward, so any two trending series correlate. After adjusting prices for Turkish CPI, removing the shared trend, and validating forecasts properly, the data does not support a strong fare effect, and the fuel-price effect is weak and not robust. Ridership growth is mainly a time trend interrupted by COVID-19.

1. Motivation

As a resident of Istanbul and regular metro user, I wanted to understand which external factors influence public transport usage. While living in the Netherlands I saw that fuel price changes were sometimes met with fare adjustments meant to steer commuter behavior, which raised the question of whether similar patterns appear in Istanbul. The project also let me practice the full data science workflow: data collection, cleaning, statistical testing, modelling and critical evaluation of my own results.

2. Hypotheses

All tests use a 5% significance level and one-sided alternatives that match the direction of each hypothesis:

- **H1:** higher fuel prices lead to higher metro ridership (H0: no effect).
- **H2:** higher metro fares lead to lower ridership (H0: no effect).
- **H3:** higher car sales lead to lower ridership (H0: no effect).

Because H2 and H3 are one-sided, a positive correlation is evidence against the alternative, and the correct conclusion is "fail to reject H0" rather than a significant effect.

3. Data

- **Ridership:** annual metro passengers, Metro Istanbul / IBB.
- **Fuel prices:** unleaded petrol price changes (Aytemiz, TPPD).
- **Metro fares:** full Istanbulkart fare history (IBB, tuhim.ibb.gov.tr). The 2013 value covers 28 June to 31 December only.
- **Car sales:** annual automobile sales from ODMD (Turkey-wide, not Istanbul-only).
- **Inflation:** Turkish consumer price inflation, annual averages (World Bank WDI, FP.CPI.TOTL.ZG, derived from TÜİK), chained into a CPI index with 2013 = 100.

Yearly fuel and fare prices are time-weighted: each price counts for the number of days it was in force, instead of taking a plain mean of the rows in the price list. Real prices are nominal prices divided by the CPI index. Population data used in the first version is not available anymore and is excluded.

4. Why inflation matters

Between 2013 and 2024 the nominal gasoline price rose about 9 times and the fare about 10 times, largely because of inflation. In real 2013 liras, gasoline moved between about 3.0 and 4.8 TL per litre with no sustained increase, and the fare fell from 1.95 to roughly 1.8 TL. Ridership grew steadily (except in 2020 to 2021), so correlating nominal prices with ridership mostly measures two trends moving together.

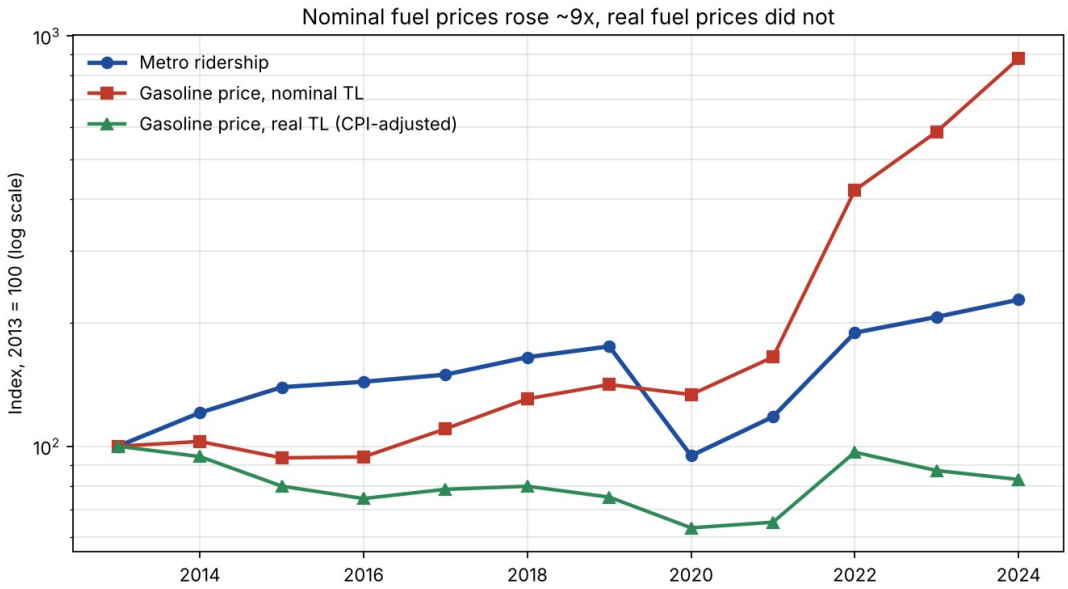


Figure 1. Ridership, nominal gasoline price and real (CPI-adjusted) gasoline price, indexed to 2013 = 100, log scale.

5. Hypothesis tests

Each relationship was tested three ways: on nominal levels (to reproduce the first version), on real levels, and on year-over-year log changes, which removes shared trends (n = 11).

Test	Fuel price	Metro fare	Car sales
Nominal levels (n = 12), Pearson r	0.79 (p = 0.002)	0.79 (p = 0.002)	-0.03 (p = 0.92)
Real levels (n = 12), Pearson r	0.22 (p = 0.50)	-0.16 (p = 0.62)	not applicable
Year-over-year change (n = 11), r	0.65 (1-sided p = 0.015)	-0.36 (1-sided p = 0.14)	-0.52 (1-sided p = 0.050)
Year-over-year, Spearman	0.52 (p = 0.10)	-0.21 (p = 0.54)	-0.28 (p = 0.40)

Multiple regression of ln(ridership) on ln(real fuel price), ln(real fare), a linear time trend and a COVID-19 dummy (2020 to 2021), n = 12, 7 residual degrees of freedom, $R^2 = 0.97$:

Term	Coefficient	p-value
ln real fuel price	-0.43	0.066
ln real fare	-0.16	0.50
Time trend (per year)	+0.061 (about 6% per year)	< 0.001
COVID-19 (2020 to 2021)	-0.68	< 0.001

Real fuel and real fare are moderately collinear (r = 0.58), which makes their separate effects hard to identify with so few observations.

Interpretation

- **H1 (fuel):** the nominal correlation disappears with real prices. A significant association remains only for year-over-year changes, and it is not robust (not significant under Spearman, and negative in the model with trend and COVID). I treat it as weak, suggestive evidence at most.
- **H2 (fare):** no evidence that higher real fares reduce ridership; fail to reject H0. The positive nominal correlation reflects inflation, not an effect of fares.
- **H3 (car sales):** borderline negative association in changes (p = 0.050), no relationship in levels. Car sales are national, ridership is for Istanbul.

6. Forecasting models

The first version reported R^2 of 0.80 to 0.82 for random forest and XGBoost. Those scores came from a random 80/20 split of only 9 usable years, i.e. 2 test rows, which is not a meaningful estimate. The revised analysis evaluates models with one-year-ahead expanding-window validation (predicting each year from 2018 to 2024 using only earlier years) and leave-one-year-out validation, and compares them with simple baselines. Random forest and XGBoost (depth 2) use real prices; the linear models use real prices plus a trend, or a trend only.

Model	One-year-ahead RMSE (M)	MAPE	Leave-one-year-out RMSE (M)
Naive: last year's value	174.3	25.0%	143.3
Linear trend only	194.2	28.3%	142.7
XGBoost (real prices)	197.7	28.3%	131.4
Random forest (real prices)	199.9	29.6%	121.7
Linear (real prices + trend)	336.9	39.8%	155.5

No model beats the naive baseline in realistic one-year-ahead forecasting; the 2020 COVID-19 collapse is a major source of error. Random forest is best under leave-one-year-out validation, but that setup is optimistic because later years help predict earlier ones. Tree-based models also cannot extrapolate beyond the training range, and the 2024 prices lie far outside earlier nominal values. For these reasons the 2025 scenario forecasts of the first version were removed: they used inputs outside the training range, and all scenarios returned nearly identical values because the model ignored the changed inputs.

7. Limitations

- Only 11 to 12 annual observations; results are exploratory and not causal estimates.
- COVID-19 years are strong outliers; the dummy handles this only crudely.
- Fuel prices and car sales are national figures while ridership is Istanbul-specific.
- Network expansion (new metro lines) and population growth are not controlled for and likely drive the upward trend.
- Real fuel and fare prices are collinear.
- CPI is an annual average, so within-year price movements are only approximated.

8. Conclusion and future work

After adjusting for inflation and trend, the data does not show that fare policy is the main driver of metro ridership, and fuel prices have at most a weak, non-robust association. The earlier conclusion that fare is the most significant factor came from comparing nominal, trending series and from evaluating models on

two test rows. Metro ridership in Istanbul over 2013 to 2024 is explained mostly by a steady upward trend, likely reflecting network expansion and population growth, with a sharp COVID-19 interruption.

Natural next steps: monthly or weekly ridership data to increase sample size, controls for network length and population, a COVID-aware model, and fare data by passenger category.

Reproducibility

All numbers in this report are produced by analysis_revised.py in the project repository, which reads the csv folder and writes the results, the annual dataset and Figure 1 to revised_output.